

 AI DEVOPS VALUE STREAM PORTAL

Enterprise Autonomous AI Agent Farm

The Universal Platform for Building MCP Servers & Deploying Autonomous AI Agents On-The-Fly

Build MCP Servers

On-the-Fly for ANY Cloud/API

Deploy AI Agents

On-the-Fly across SDLC

98% MTTR Reduction

From 45-90 min to 8 sec

Zero Duplicate ITSM

Stateful SHA-256 Memory

The Challenge: Manual SRE & Fragmented DevOps Silos

Traditional DevOps & SRE Pain Points

-   High MTTR (45–90 min): Engineers manually dig through 300+ lines of container logs.
-  Severe Alert Fatigue: Cascading failures flood ServiceNow with 20+ duplicate tickets.
-  Disconnected Tool Silos: Docker, GitHub, Jenkins, and ITSM operate in isolation.
-   Manual Remediation: Engineers copy-paste stack traces and manually write RCA notes.

The AI DevOps Value Stream Portal Solution

-  98% MTTR Drop (5–12 sec): Autonomous background detection and deterministic RCA.
-  SHA-256 Anti-Storm Memory: Prevents duplicate incident flooding completely.
-  Universal FastMCP Gateway (:5001): Standardized JSON-RPC proxy connecting all tools.
-  Closed-Loop Auto-Resolution: Enriches ServiceNow work notes & marks tickets Resolved.

The Dual-Engine Foundation: Servers & Agents On-The-Fly

🔗 Engine 1: Build MCP Servers On-The-Fly

- Prompt-Driven Synthesis: Chat with Mistral AI to build tool suites for AWS, K8s, Jira, Datadog, or custom APIs.
- Automated Schema Generation: Synthesizes tool definitions, argument types, and parameter docs automatically.
- Credential Vault Management: Auto-configures local per-server .env vaults and authentication headers.
- 1-Click Hot-Reload: Deploys instantly to the secured Gateway (:5001) with zero downtime.

🔗 Engine 2: Deploy AI Agents On-The-Fly

- Conversational Discovery: Describe any automation goal in natural language (SRE, CI/CD, Code QA, Infra).
- Live Capability Validation: Validates required tools against registered MCP servers before generating code.
- Deterministic Code Synthesis: Generates executable Python workflows (workflow.py) with zero hardcoding.
- Fleet Management: Deploys 5s–30s autonomous background watchdogs with execution telemetry.

Full-Lifecycle DevOps AI Agent Farm (CM, CI, CD, CT & AIOps)

1. CM (Continuous Management)

GitHub branch policies, JPA schema constraint gates, automated ServiceNow Change Requests.

2. CI (Continuous Integration)

Jenkins build failure triage, compilation error correlation with Git authors, auto-remediation.

3. CD (Continuous Deployment)

Post-deployment container health gates on :7000, automated clean rebuilds & rollback triggers.

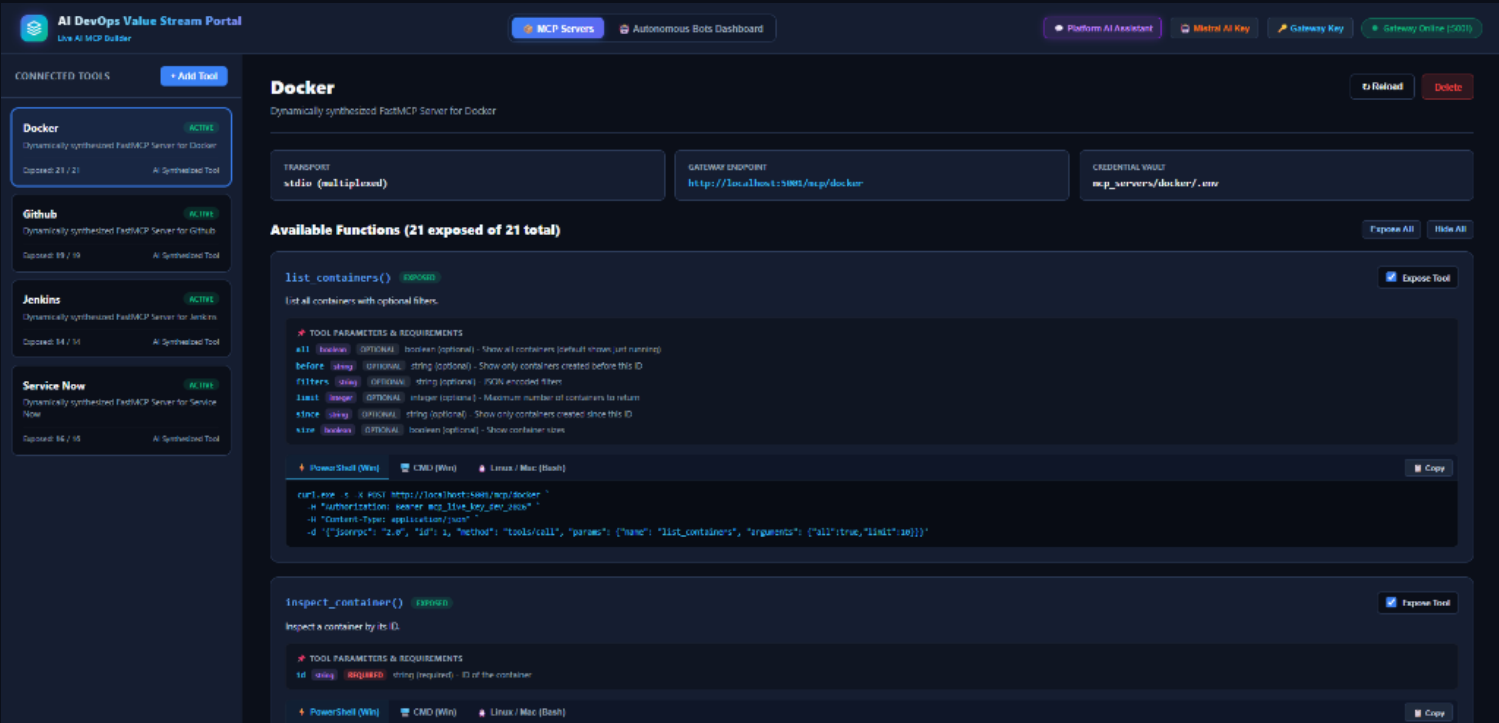
4. CT (Continuous Testing)

JUnit / integration test failure isolation, regression triage, flaky test fingerprinting.

5. AIOps (Observability)

5s container log watchers, SHA-256 deduplication, Mistral RCA & ServiceNow auto-resolution.

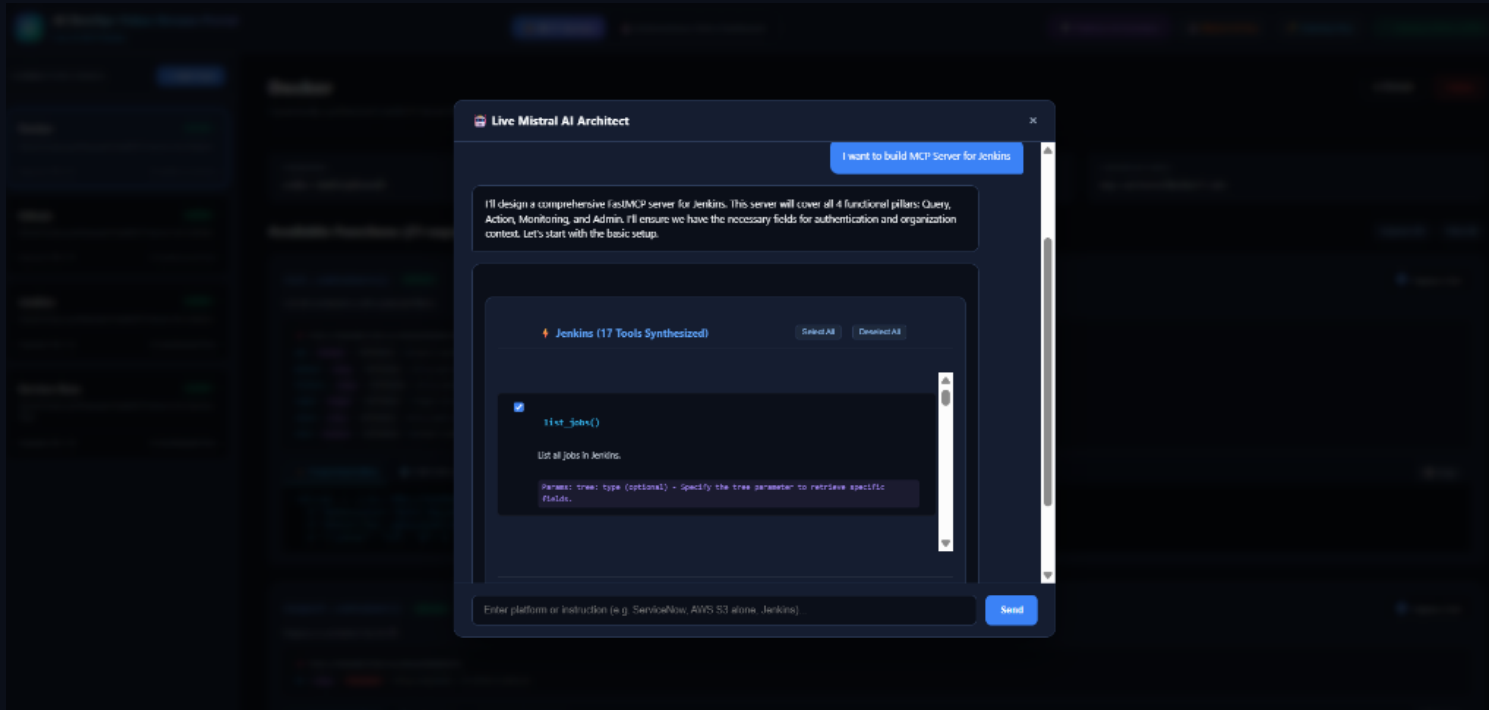
Platform Tour: Web Dashboard & Connected MCP Servers



Unified MCP Fleet Management

- Pre-Integrated Connectors: Docker (21 tools), GitHub (19 tools), Jenkins (14 tools), ServiceNow (16 tools).
- Live Gateway Route: `http://localhost:5001/mcp/<server>`
- Function Explorer: Real-time schema parameters, types, and instant cURL snippets for testing.
- Per-Server Credential Vault: Secure isolation with `.env` support and zero-downtime hot reloading.

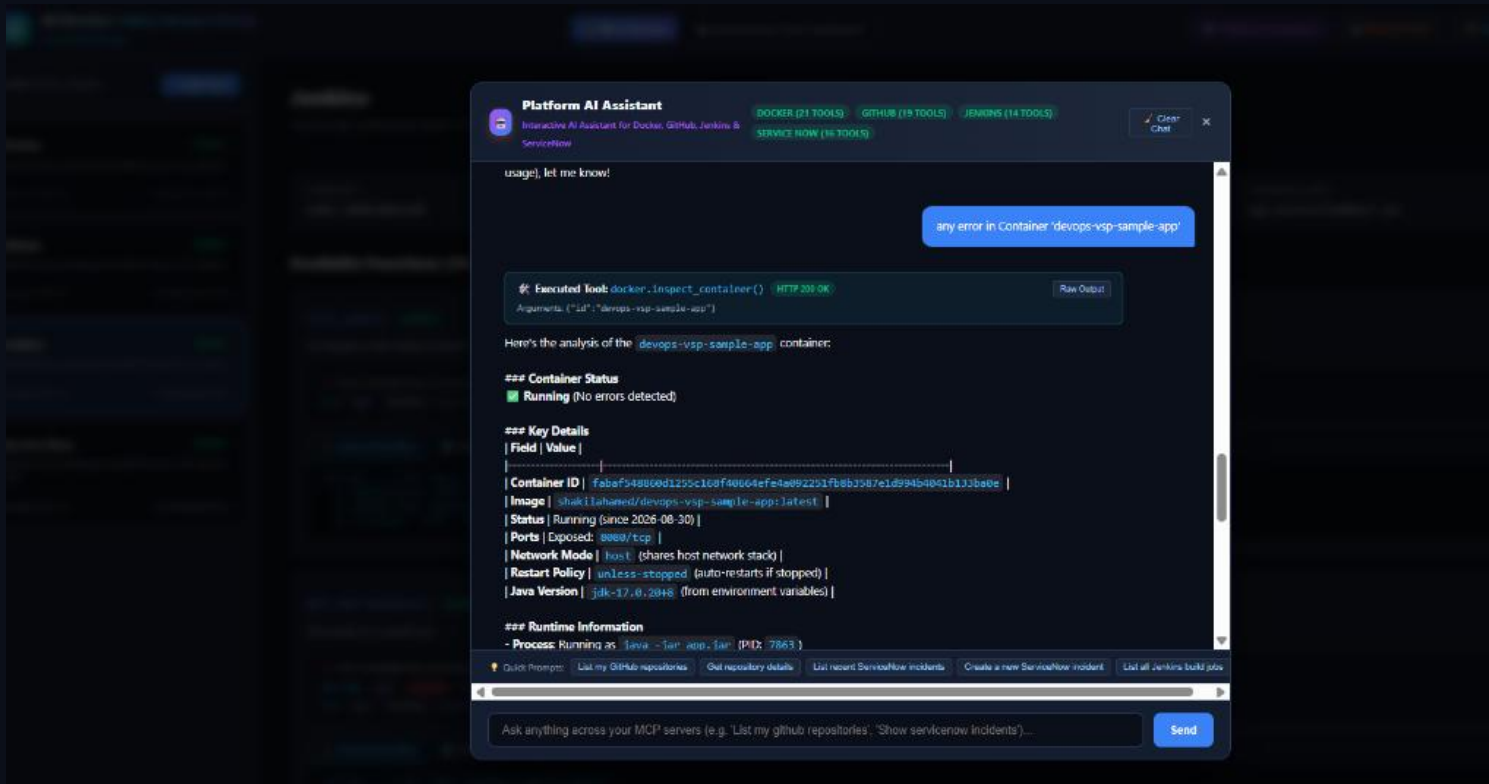
Capability Showcase: Building MCP Servers On-The-Fly



Conversational Server Architect

- Natural Language Prompting: Describe the target platform (e.g. 'Build MCP Server for Jenkins').
- Automated Tool Scaffolding: Mistral automatically synthesizes 17 tools covering Query, Action, Monitoring & Admin.
- Interactive Selection: Toggle specific tool exposures on/off before generation.
- 1-Click FastMCP Deploy: Generates Python server scripts and saves credential vaults automatically.

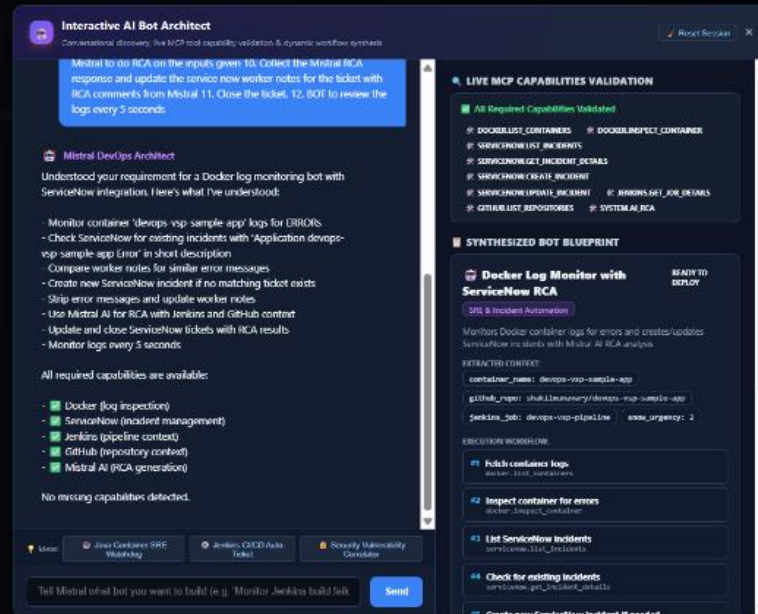
Universal Platform AI Assistant (Cross-Platform Invocations)



Live MCP Multi-Server Chat

- Cross-Platform Intelligence: Chat with an AI assistant that can invoke any registered tool in real time.
- Docker Inspections: 'Any error in Container devops-vsp-sample-app' -> returns live PID, status, and health.
- GitHub Queries: 'List my repositories' -> returns 30+ authenticated repos via GitHub PAT.
- ServiceNow & Jenkins: Query live build jobs and open ITSM incident records seamlessly.

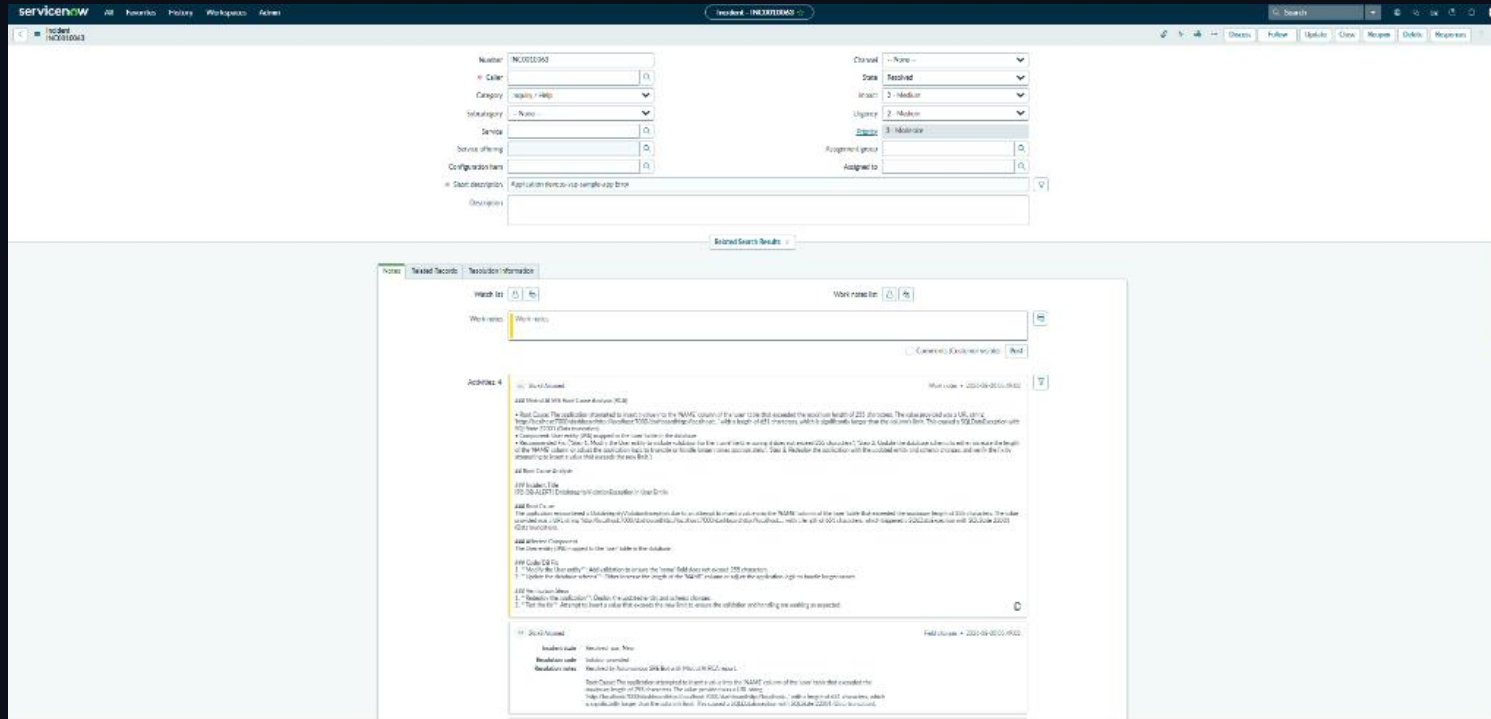
Capability Showcase: Deploying Autonomous Bots On-The-Fly



🔗 Autonomous Bot Architect

- **Conversational Discovery:** Describe the workflow (log monitoring, deduplication, AI RCA, ServiceNow ticketing).
- **Live Capability Validation:** Validates required MCP tools against Docker, ServiceNow, Jenkins & GitHub.
- **12-Step Synthesized Blueprint:** Mistral generates the complete execution sequence.
- **1-Click Fleet Deployment:** Deploys as an autonomous background watchdog with execution telemetry.

Real-World Showcase: ServiceNow Auto-Resolution with Mistral RCA



Incident INC0010063 Resolved

- Failure Detected: Spring Boot SQL 22001 column truncation error on user.name (651 chars).
- Initial Triage Notice: Automatically logs 'SRE AI agent is analyzing issue' into work_notes.
- Mistral AI RCA Report: Identifies root cause and generates exact JPA entity fix (@Size(max=255)).
- Non-Destructive Auto-Closure: Appends full RCA markdown to work_notes and marks ticket Resolved (state: 6).

Business Impact, SRE Metrics & ROI Analysis

⚡ 98% MTTR Reduction

Mean Time to Resolve drops from 45–90 min down to 5–12 seconds.

🔍 Zero Duplicate Storms

Stateful SHA-256 error fingerprinting prevents duplicate ticket creation.

💰 92% Token Cost Cut

Precision stack trace stripping isolates root exceptions from 350+ lines of logs.

💰 >\$138,000 / Year ROI

Annual savings in engineering hours and reduced downtime per 50-service fleet.

Summary & Open Source Links

Join the Autonomous DevOps Movement!

- Dual-Engine Platform: Build MCP servers on-the-fly & deploy autonomous AI agents across the entire SDLC.
- Open Source & Extensible: Model Context Protocol (JSON-RPC 2.0) + FastMCP Gateway (:5001).
- Production Ready: Deployable via Bare-Metal Daemon, Docker Compose, and Kubernetes.
- GitHub Repository: <https://github.com/shakilmunavary/ai-mcp-builder>